AUPRC-Guided Bilevel Mask Optimization for Interpretable Alzheimer's Disease Classification under Class Imbalance

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Abstract

Early-stage Alzheimer's disease (AD) classification is important for clinical intervention and prognosis assessment. However, existing graph-based methods often overlook decision interpretability and suffer from class imbalance, where accuracy-oriented optimization may fail to reflect minority-class recognition. To address these issues, we propose an AUPRC-oriented bilevel mask learning framework for imbalanced and interpretable AD classification. The lower level optimizes the graph classifier on the training set, while the upper level learns an interpretable edge-level mask on the validation set to directly improve AUPRC. Implicit gradient-based coupling enables end-to-end joint optimization of the classifier and mask without pre-freezing the main model. Experiments on OASIS and ADNI demonstrate that our method improves classification performance, especially AUPRC, and produces more stable and consistent edge-level explanations for brain connectivity analysis.

Keywords: Alzheimer's disease; Graph neural networks; Class imbalance; Bilevel optimization; Interpretable mask learning

1 Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that affects memory, cognition, and daily functioning. Since AD-related changes may begin before severe clinical symptoms appear, early-stage classification is an important problem for prognosis and potential intervention. In neuroimaging studies, the brain is often modeled as a structural or functional network, where nodes represent anatomical regions of interest (ROIs) and edges describe connectivity between these regions. Prior studies suggest that AD is not limited to isolated regional damage, but is associated with distributed disruption across large-scale brain networks [1, 17]. This motivates the use of graph-based machine learning methods that can model relationships among brain regions rather than treating each region independently.

Diffusion tensor imaging (DTI) provides a way to reconstruct white-matter structural connectivity between brain regions [2]. After preprocessing and tractography, each subject can be represented as a weighted brain graph, where edge weights describe estimated structural connection strengths. Compared with conventional statistical analysis or handcrafted graph measures, graph neural networks (GNNs) provide a more flexible framework for learning representations directly from graph-structured connectome data. Recent work has explored GNNs, graph attention models, and transformer style architectures for brain disorder analysis and AD classification [5, 11, 24, 25].

However, connectome-based AD classification faces two practical challenges. First, neuroimaging datasets are often imbalanced across disease stages. Early or transitional groups, such as subjective memory concern (SMC) or mild cognitive impairment (MCI), may contain fewer samples than cognitively normal controls or more advanced AD groups. In this setting, a model trained with standard cross-entropy loss may achieve acceptable overall accuracy while still performing poorly on clinically important minority classes. Therefore, evaluation metrics such as the area under the precision–recall curve (AUPRC) are often more informative than accuracy or AUROC when the positive class is rare [7, 16]. This project therefore treats AUPRC as a key metric for evaluating imbalanced AD classification.

Second, interpretability is important in medical applications. Although GNN-based models can learn complex graph representations, their predictions are often difficult to interpret. For AD classification, it is not sufficient to only predict the disease stage; we also want to understand which brain connections contribute to the model's decision. Existing graph explanation methods, such as GNNExplainer, identify important nodes or edges after a model has already been trained [23]. Brain-network-specific explanation methods have also been proposed to identify disease-related subnetworks [6]. However, post-hoc explanations may be unstable or not fully aligned with the classifier's training objective.

In this class project, we study an interpretable graph-learning pipeline for imbalanced AD classification using structural brain networks. The project focuses on three questions: (1) how to represent DTI-based connectomes as graphs suitable for GNN classification, (2) how AUPRC-oriented learning and evaluation can better reflect performance under class imbalance, and (3) how edge-level importance masks can provide interpretable connectivity patterns related to AD progression. Rather than presenting the system as a new clinical diagnostic tool, our goal is to implement, evaluate,

and analyze graph-based learning strategies for imbalanced and interpretable brain-network classification.

2 Related Work

2.1 Connectome-based Alzheimer's Disease Analysis

Brain connectivity analysis has been widely used to study AD and related cognitive disorders. Instead of focusing only on individual brain regions, connectome-based analysis models the brain as a network of interacting regions. Prior studies have shown that AD is associated with disruptions in large-scale brain networks, including default-mode and memory-related circuits [1, 9, 17]. Structural connectomes reconstructed from DTI can capture white-matter connectivity patterns, providing useful information for studying disease progression [2, 15]. These findings motivate machine learning methods that can use network-level information for AD stage classification.

2.2 Graph Neural Networks for Brain Network Classification

Graph neural networks have become a natural choice for learning from brain networks because they operate directly on graph-structured data. In a connectome graph, each ROI can be treated as a node, and structural or functional connectivity can be treated as weighted edges. Compared with traditional feature-based models, GNNs can aggregate information from neighboring regions and learn topology-aware graph representations. Several recent methods have applied graph neural networks or graph attention mechanisms to brain disorder classification [5, 25]. Transformer-style models have also been adapted to brain networks to capture long-range dependencies among ROIs [11, 24]. These methods show the potential of graph learning for neuroimaging analysis, but they may still be sensitive to data imbalance and may require additional mechanisms for interpretability.

2.3 Learning under Class Imbalance

Class imbalance is common in medical datasets because some disease stages or clinical subgroups may have far fewer samples than others. In AD classification, early or transitional stages are especially important but can be underrepresented. Standard accuracy-driven training may favor the majority class and fail to detect minority cases. Precision–recall based metrics are often preferred in imbalanced binary classification because they focus more directly on the positive class performance [7, 16]. AUPRC summarizes the trade-off between precision and recall over different thresholds and is therefore useful when the positive class is rare. Recent optimization work has also studied differentiable surrogates for directly optimizing areas under precision–recall curves [14]. In this project, we use AUPRC as a primary evaluation criterion and consider training objectives that better align with this metric.

2.4 Interpretability for Graph-based Medical Models

Interpretability is essential for medical machine learning because model predictions should be connected to clinically meaningful evidence. For graph learning, explanation methods often aim to identify important nodes, edges, or subnetworks. GNNExplainer is a general method that explains graph neural network predictions by learning compact subgraph masks [23]. In brain-network analysis, BrainNNExplainer extends this idea to identify salient brain connections associated with disease classification [5, 6]. These methods provide useful tools for interpreting GNN predictions, but many of them are post-hoc: the classifier is trained first, and explanations are extracted afterward. In this project, we focus on edge-level interpretability for structural connectomes and analyze whether learned connectivity masks can highlight plausible disease-related subnetworks.

3 Preliminaries

3.1 Structural Brain Graph Representation

Each subject is represented as a weighted brain graph

$$G = (V, W, X),$$

where $V = \{v_1, v_2, \ldots, v_N\}$ is the set of brain regions of interest, $W \in \mathbb{R}^{N\times N}$ is the weighted connectivity matrix, and $X \in \mathbb{R}^{N\times d}$ is the node feature matrix. The entry w_{ij} in W represents the structural connection strength between ROI v_i and ROI v_j. In DTI-based structural connectomes, these edge weights are commonly derived from tractography-based estimates of white-matter connectivity [2]. Since structural brain graphs are usually undirected, W is typically symmetric.

In many connectome datasets, raw node attributes are limited or unavailable. Therefore, node features can be constructed using ROI identity features, local graph statistics, or other subject-invariant descriptors. For example, a simple option is to combine one-hot ROI identifiers with degree-based local structural information. Local degree profile features summarize the degree of each node and the degree statistics of its neighbors, providing a lightweight way to describe local graph structure [4].

3.2 Graph Neural Network Classifier

Given a brain graph G, a graph neural network learns node representations by repeatedly aggregating information from neighboring nodes. A generic message-passing layer can be written as

$$h_i^{(k+1)} = \phi\left(h_i^{(k)}, \sum_{j\in\mathcal{N}(i)} \psi(h_i^{(k)}, h_j^{(k)}, w_{ij})\right),$$

where $h_i^{(k)}$ is the representation of node v_i at layer k, $\mathcal{N}(i)$ denotes the neighbors of v_i, w_{ij} is the edge weight, and ϕ and ψ are learnable functions. After several graph layers, a graph-level representation can be obtained using a readout

function such as sum pooling, mean pooling, or attention pooling:

$$h_G = \mathrm{READOUT}\left(\{h_i^{(K)} : v_i \in V\}\right).$$

The graph representation h_G is then passed to a multilayer perceptron to predict the class label, such as cognitively normal, MCI, or AD.

3.3 AUPRC for Imbalanced Classification

For binary classification, precision and recall are defined as

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \qquad \mathrm{Recall} = \frac{TP}{TP + FN},$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. The precision–recall curve evaluates the trade-off between precision and recall as the decision threshold changes. The area under this curve, AUPRC, is especially useful when the positive class is rare because it directly reflects the model's ability to retrieve positive samples while controlling false positives [7, 16].

In imbalanced AD classification, AUPRC can be more informative than accuracy. A classifier may achieve high accuracy by predicting most samples as the majority class, but such behavior may miss clinically important minority cases. Therefore, this project reports AUPRC as a primary metric and AUROC as a complementary metric.

3.4 Edge Masks for Interpretability

To interpret which brain connections are important for classification, an edge mask can be applied to the connectivity matrix. Let $M \in [0,1]^{N\times N}$ denote a soft edge-importance mask. The masked connectivity matrix is

$$W' = W \odot M,$$

where $\odot$ denotes element-wise multiplication. A larger value of m_{ij} indicates that the connection between ROI v_i and ROI v_j is considered more important by the model. After training, the top-ranked edges in M can be visualized as discriminative subnetworks. This provides a way to connect model predictions with brain connectivity patterns and supports qualitative analysis of disease-related structural changes.

4 Methods

4.1 Overview of AOBML

Building on the graph representation and edge-mask formulation introduced in the preliminaries, we propose an AUPRC-Oriented Bilevel Mask Learning framework, termed AOBML, for imbalanced and interpretable AD classification. The framework jointly optimizes two components: a graph classifier f_θ for disease-stage prediction and a global edge-mask parameter $H \in \mathbb{R}^{N\times N}$ for identifying discriminative structural connections. Unlike post-hoc explanation methods that generate importance scores after classifier training, AOBML integrates mask learning into the training process

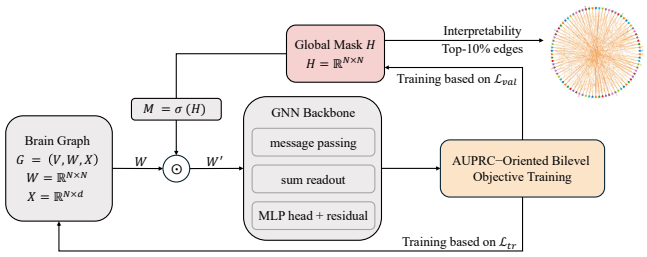

Figure 1. Overview of AOBML. The learnable mask logits H are converted into a soft edge mask and applied to the structural connectivity matrix W. The masked graph is then processed by a GNN classifier. The backbone parameters and mask logits are optimized through an AUPRC-oriented bilevel objective, where the lower level updates the classifier and the upper level updates the interpretable mask.

itself, allowing the learned mask to be directly aligned with the classification objective.

Given a structural connectivity matrix W, the learnable mask logits H are transformed into a soft mask by an element-wise sigmoid function. The masked graph is then constructed by reweighting the original connectivity matrix before message passing. The classifier is trained on the training set, while the mask is updated according to validation performance. In this way, AOBML encourages the mask to preserve connections that generalize beyond the training data and contribute to minority-class recognition under imbalanced distributions.

4.2 Mask-Conditioned Graph Classification

For each subject, we first obtain a soft edge mask

$$M = \sigma(H),$$

where $\sigma(\cdot)$ is applied element-wise. Since structural brain connectivity is undirected and self-connections are not considered, we impose symmetry and remove diagonal entries after each mask update:

$$H \leftarrow \frac{1}{2}(H + H^\top), \qquad \mathrm{diag}(H) \leftarrow \mathbf{0}. \tag{1}$$

The mask is applied to the structural connectivity matrix as

$$\widetilde{W} = W \odot M, \tag{2}$$

where $\odot$ denotes element-wise multiplication. The masked connectivity matrix $\widetilde{W}$ is then used in the edge-weight-aware GNN backbone. Specifically, the message passing operation uses the masked edge weight $\widetilde{w}_{ij}$ instead of the original connection strength w_{ij}, so that irrelevant or noisy connections can be suppressed during representation learning.

After several graph layers, a graph-level embedding is obtained through sum pooling and passed into an MLP classifier. The final prediction logits are denoted as

$$\mathbf{z} = f_\theta(G; H),$$

where θ represents the classifier parameters and H controls the structural mask used by the classifier.

4.3 AUPRC-Oriented Bilevel Learning

AOBML formulates classifier training and mask learning as a bilevel optimization problem. The lower level optimizes the classifier parameters on the training set, while the upper level updates the mask using validation-set performance:

$$\min_{H} \ \mathcal{L}_{\text{val}}(\theta^*(H), H) \quad \text{s.t.} \quad \theta^*(H) = \arg\min_{\theta} \mathcal{L}_{\text{tr}}(\theta, H). \tag{3}$$

Here, $\mathcal{L}_{\text{tr}}$ denotes the lower-level training objective, and $\mathcal{L}_{\text{val}}$ denotes the upper-level validation objective. This design has two advantages. First, the classifier is optimized to learn discriminative graph representations from the masked connectome. Second, the mask is selected according to validation performance, which reduces the risk of learning explanations that only fit the training data. Therefore, the learned mask is encouraged to be both predictive and generalizable.

4.4 AUPRC Surrogate Loss

Since AUPRC is based on ranking operations and is not directly differentiable, we adopt the SOAP loss as a differentiable surrogate for average precision [?]. For a mini-batch b, let $b^+ = \{i \in b \mid y_i = 1\}$ denote the set of positive samples. Given logits $\mathbf{z}_i$, the positive-class score is computed as

$$s_i = \text{softmax}(\mathbf{z}_i)_1.$$

For each positive anchor $i \in b^+$, SOAP estimates a soft precision by replacing the discrete ranking indicator with a smooth surrogate. We use the squared-hinge surrogate

$$\ell(s_j, s_i) = \left[\max\{m - (s_i - s_j), 0\}\right]^2, \tag{4}$$

where m is the margin. The differentiable AP surrogate is then written as

$$\widetilde{\text{AP}} = \frac{1}{|b^+|} \sum_{i \in b^+} \frac{\sum_{j \in b^+} \ell(s_j, s_i)}{\sum_{j \in b} \ell(s_j, s_i)}. \tag{5}$$

The SOAP loss is defined as

$$\mathcal{L}_{\text{SOAP}} = 1 - \widetilde{\text{AP}}. \tag{6}$$

Accordingly, the lower-level objective is

$$\mathcal{L}_{\text{tr}}(\theta, H) = \mathcal{L}_{\text{SOAP}}(\mathcal{D}_{\text{tr}}; \theta, H). \tag{7}$$

4.5 Mask Regularization and Upper-Level Objective

To obtain interpretable and stable edge masks, we regularize the sigmoid-transformed mask $M = \sigma(H)$. First, a sparsity penalty is used to discourage overly dense masks:

$$\mathcal{L}_s(M) = \frac{1}{N(N-1)} \sum_{i \neq j} m_{ij}. \tag{8}$$

Second, an entropy penalty is introduced to encourage confident mask values close to either 0 or 1:

$$\mathcal{L}_e(M) = \frac{1}{N(N-1)} \sum_{i \neq j} \mathcal{H}(m_{ij}), \tag{9}$$

where $\mathcal{H}(m) = -m \log(m + \epsilon) - (1 - m) \log(1 - m + \epsilon)$, and ϵ is a small constant for numerical stability.

The upper-level objective is therefore defined as

$$\mathcal{L}_{\text{val}}(\theta, H) = \mathcal{L}_{\text{SOAP}}(\mathcal{D}_{\text{val}}; \theta, H) + \lambda_s \mathcal{L}_s(M) + \lambda_e \mathcal{L}_e(M), \tag{10}$$

where λ_s and λ_e control the strengths of sparsity and entropy regularization. By optimizing this objective on validation data, the mask is guided toward structural connections that improve AUPRC while remaining compact and interpretable.

4.6 First-Order Bilevel Optimization with Muon

Directly solving Eq. (3) requires differentiating through the lower-level optimum, which can be computationally expensive. To avoid explicit second-order derivatives, we adopt a first-order minimax reformulation:

$$\min_{H,\theta} \max_{z} \ \mathcal{L}_{\text{val}}(\theta, H) + \frac{1}{\rho} \left[\mathcal{L}_{\text{tr}}(\theta, H) - \mathcal{L}_{\text{tr}}(z, H)\right], \tag{11}$$

where z is an auxiliary copy used to approximate the lower-level solution, and $\rho > 0$ is a penalty coefficient. This reformulation allows the model to update H, θ, and z using only first-order gradients.

For the mask parameter H, the gradient direction is computed as

$$g_H = \nabla_H \mathcal{L}_{\text{val}}(\theta, H) + \frac{1}{\rho} \left[\nabla_H \mathcal{L}_{\text{tr}}(\theta, H) - \nabla_H \mathcal{L}_{\text{tr}}(z, H)\right]. \tag{12}$$

To improve the stability of matrix-shaped updates, we employ the Muon optimizer [10]. Specifically, for the mask matrix, we maintain a momentum estimate

$$P_t = (1 - \alpha) P_{t-1} + \alpha g_H, \tag{13}$$

where $\alpha \in (0, 1)$ is the momentum coefficient. The momentum matrix is then orthogonalized using Newton–Schulz iterations to obtain an update direction O_t. The mask logits are updated by

$$H_{t+1} = H_t - \eta_H O_t, \tag{14}$$

where η_H is the learning rate for the mask. After each update, the symmetry and zero-diagonal constraints in Eq. (1) are applied again.

The classifier parameters θ and auxiliary variable z are updated in the same first-order optimization loop. As a result, AOBML provides a single-stage training procedure that jointly learns the disease classifier and the interpretable structural mask without requiring a separately pre-trained or frozen backbone.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate the proposed AOBML framework on two public neuroimaging datasets for Alzheimer's disease analysis. The first dataset is from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [13]. It contains 154 cognitively normal (CN) subjects, 120 subjects with subjective memory concern (SMC), and 67 subjects with mild cognitive impairment (MCI). The mean ages of the three groups are 77.46 ± 5.97, 75.87 ± 5.11, and 73.80 ± 5.95, respectively, with 95, 49, and 55 women in each group.

The second dataset is derived from the Open Access Series of Imaging Studies-3 (OASIS-3) [12]. It includes 806 CN subjects, 150 MCI subjects, and 48 AD subjects. Their mean ages are 68.90 ± 8.99, 75.31 ± 6.81, and 74.71 ± 8.60, respectively, with 503, 70, and 22 women in the corresponding groups.

For both datasets, structural connectivity networks are reconstructed using the FSL probabilistic tractography pipeline, including Bedpostx and Probtrackx [2]. For ADNI, the structural network of each subject is represented as an 82×82 matrix based on 82 cortical and subcortical ROIs defined by FreeSurfer v6.0 [8]. For OASIS, each subject is represented by a 132×132 structural connectivity matrix constructed from ROIs defined by the Harvard–Oxford atlas [18] and the Automated Anatomical Labeling (AAL) atlas [20]. These two datasets use different parcellation schemes and graph resolutions, which allows us to examine the robustness of the proposed method across different connectome construction settings.

Baselines. We compare AOBML with five representative graph-based methods. **BasicGNN** uses the same GNN backbone as our method but is trained with standard supervised learning and does not include bilevel mask learning. **BrainNNExplainer** [5] is an explainability-oriented brain graph model that identifies salient brain connections for prediction. **BGAN** [25] adopts graph attention mechanisms for AD classification. **BrainNETTF** [11] introduces a Transformer-style architecture for brain network representation learning. **ALTER** [24] models both local and long-range ROI dependencies by combining adaptive biased random walks with self-attention. All compared methods are evaluated using the same data splits, preprocessing pipeline, and evaluation protocol.

Implementation Details. All methods are implemented in PyTorch with the random seed fixed to 42. We adopt 5-fold stratified cross-validation and report the mean and standard deviation across folds. The input to the model is a subject-level normalized structural connectivity tensor, denoted as $\texttt{normsc} \in \mathbb{R}^{N \times R \times R}$, where N is the number of subjects and R is the number of ROIs.

The backbone network contains two graph layers followed by an MLP classifier with hidden dimension 16. Sum pooling is used as the graph-level readout function, and the dropout rate is set to 0.25. For AOBML, each fold is trained for 200 epochs. We use Adam with learning rate 1×10^{-3} for the first 100 epochs and 5×10^{-4} for the remaining epochs. The batch size is set to 16. In the bilevel optimization process, exponential moving average smoothing is applied with coefficient $\beta = 0.9$, and the penalty parameter is set to $\rho = 5$. The checkpoint with the highest validation AUPRC is selected for testing. For SOAP optimization, we use a class-balanced sampler with $B_{\text{pos}} = 2$. The sparsity and entropy regularization weights are set to $\lambda_s = 5 \times 10^{-3}$ and $\lambda_e = 10^{-3}$, respectively. For methods that explicitly learn explanation masks, we average the learned masks across folds to obtain more stable estimates of discriminative structural connections.

Table 1. Classification performance on ADNI for NC vs. SMC and SMC vs. MCI.

Method	NC vs. SMC		SMC vs. MCI	
	AUPRC(%)	AUROC(%)	AUPRC(%)	AUROC(%)
BasicGNN	34.82±4.13	52.44±8.30	74.45±6.51	60.43±10.85
BrainNNExplainer	42.05±6.85	60.28±4.57	74.11±7.39	59.38±11.93
BGAN	38.89±6.46	53.69±6.83	67.31±3.40	50.63±5.97
BrainNETTF	39.12±8.26	55.65±7.90	73.37±6.22	59.63±11.47
ALTER	37.67±9.13	53.79±9.18	68.29±6.31	51.69±10.20
AOBML (Ours)	**42.96±7.61**	**61.81±6.74**	**77.67±3.28**	**61.72±3.74**

Table 2. Classification performance on OASIS for NC vs. MCI and MCI vs. AD.

Method	NC vs. MCI		MCI vs. AD	
	AUPRC(%)	AUROC(%)	AUPRC(%)	AUROC(%)
BasicGNN	22.05±3.50	57.18±4.21	28.44±4.57	53.67±7.62
BrainNNExplainer	24.29±4.27	57.97±3.94	33.84±10.97	51.64±13.47
BGAN	22.78±8.07	56.08±3.72	34.29±10.08	52.48±11.02
BrainNETTF	26.16±7.27	63.49±6.79	26.65±4.35	49.23±10.19
ALTER	25.43±2.30	**64.84±5.47**	33.88±5.05	**56.33±10.81**
AOBML (Ours)	**27.31±5.86**	64.11±4.47	**35.33±10.07**	55.13±10.45

5.2 Results

Due to the pronounced class imbalance in the binary classification tasks, we use AUPRC as the primary evaluation metric and report AUROC as a complementary measure of overall class separability.

Table 1 presents the results on ADNI. AOBML achieves the best performance on both NC vs. SMC and SMC vs. MCI. For NC vs. SMC, AOBML obtains 42.96 ± 7.61 AUPRC and 61.81 ± 6.74 AUROC, outperforming the strongest baseline by 0.91 percentage points in AUPRC and 1.53 percentage points in AUROC. For the more challenging SMC vs. MCI task, AOBML reaches 77.67 ± 3.28 AUPRC and 61.72 ± 3.74 AUROC. Compared with the best competing method, this corresponds to improvements of 3.22 percentage points in

AUPRC and 1.29 percentage points in AUROC. The relatively small standard deviation in AUPRC also suggests that AOBML provides more stable cross-fold performance on this task.

The results on OASIS are reported in Table 2. On NC vs. MCI, AOBML achieves the highest AUPRC of 27.31 ± 5.86 while maintaining competitive AUROC. On MCI vs. AD, where the AD class contains substantially fewer samples, AOBML again achieves the best AUPRC of 35.33 ± 10.07 and obtains AUROC comparable to the strongest baseline. Overall, AOBML achieves the highest AUPRC across all four binary classification tasks on ADNI and OASIS. These results indicate that the proposed AUPRC-oriented bilevel mask learning strategy can improve minority-class recognition while preserving competitive overall discrimination ability.

5.3 Biological Insights and Model Interpretability

We further investigate the discriminative structural connections identified by AOBML. The top-ranked subnetworks are visualized in Fig. 2. For the ADNI NC vs. SMC task, AOBML highlights a distributed subnetwork involving the bilateral hippocampus and amygdala, together with their connections to frontal and temporal cortices. This pattern suggests that medial temporal–limbic connectivity alterations may emerge at the subjective memory concern stage, which is consistent with previous findings on hippocampal disconnection in preclinical AD [15, 21]. The additional involvement of insular and frontal regions indicates that early connectivity changes may extend beyond memory-related circuits [22].

For the ADNI SMC vs. MCI task, the selected edges shift toward the bilateral posterior cingulate cortex, precuneus, superior frontal cortex, and superior parietal regions. These areas are key hubs of the default mode and fronto-parietal networks and have been widely reported to be vulnerable during the MCI stage [3, 9]. This transition from medial temporal–limbic regions to broader association networks suggests a progressive pattern of structural network disruption along the AD continuum [17, 19]. Similar patterns are also observed on OASIS, supporting the consistency of the learned masks across datasets.

6 Conclusion

In this work, we proposed AOBML, an AUPRC-oriented bilevel mask learning framework for imbalanced and interpretable Alzheimer's disease classification. Different from post-hoc explanation methods, AOBML learns the classifier and the structural connectivity mask within a unified optimization process. By introducing an AUPRC-driven validation objective, the framework is better aligned with minority-class recognition under imbalanced data distributions. Meanwhile, the learned edge-level mask provides interpretable structural subnetworks associated with AD progression.

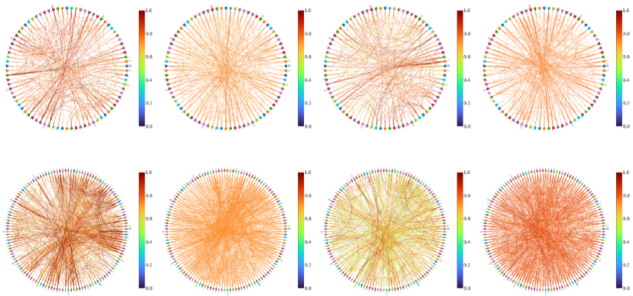

Figure 2. Visualization comparison of discriminative structural subnetworks identified by BrainNNExplainer and the proposed AOBML across AD-related classification tasks. The columns correspond to ADNI NC vs. SMC, ADNI SMC vs. MCI, OASIS NC vs. MCI, and OASIS MCI vs. AD, from left to right. The first row presents the BrainNNExplainer baseline results, whereas the second row presents the subnetworks identified by AOBML.

Experiments on ADNI and OASIS demonstrate that AOBML consistently improves AUPRC across multiple binary classification tasks while maintaining competitive AUROC performance. The identified subnetworks also show biologically meaningful patterns, including early medial temporal–limbic alterations and later involvement of default mode and fronto-parietal regions. These results suggest that AOBML can serve as an effective framework for robust and interpretable connectome-based AD analysis.

Acknowledgments

For privacy and academic integrity considerations, and because we plan to conduct further research and attempt to publish this work, we respectfully request that this report and its accompanying materials not be posted or shared publicly online.

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